# Exerciții

① $\sum\limits_{n\geq1}\frac{\alpha^n}{n^\beta\cdot\sin\frac1n}$

Crit. Rap. $\Rightarrow \lim\limits_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim\limits_{n\to\infty}\frac{\alpha^{n+1}}{(n+1)^\beta\sin\frac{1}{n+1}}\cdot\frac{n^\beta\sin\frac1n}{\alpha^n}$

$=\lim\limits_{n\to\infty}\alpha\cdot\left(\frac{n}{n+1}\right)^\beta\cdot\frac{\sin\frac1n}{\sin\frac1{n+1}}=\lim\limits_{n\to\infty}\alpha\cdot\underset{\downarrow\atop 1}{\left(\frac{n}{n+1}\right)^{\beta-1}}\cdot\underset{\downarrow\atop 1}{\frac{\sin\frac1n}{\frac1n}}\cdot\underset{\downarrow\atop 1}{\frac{\frac1{n+1}}{\sin\frac1{n+1}}}$

$\lim\limits_{n\to\infty}\frac{x_{n+1}}{x_n}=\alpha$

I Dacă $\alpha<1$ atunci $\sum x_n$ convergent

II Dacă $\alpha>1$ atunci $\sum x_n$ divergent

III Dacă $\alpha=1$ atunci $x_n=\frac{1}{n^\beta\sin\frac1n}$

fie $y_n=\frac{1}{n^{\beta-1}}$

Crit. la limită $\Rightarrow \lim\limits_{n\to\infty}\frac{x_n}{y_n}=\lim\limits_{n\to\infty}\frac{n^{\beta-1}}{n^\beta\sin\frac1n}=$

$=\lim\limits_{n\to\infty}\frac{1}{n\sin\frac1n}=\lim\limits_{n\to\infty}\frac{1}{\frac{\sin\frac1n}{\frac1n}}=1\in(0,+\infty)$

$\Rightarrow\sum x_n\sim\sum y_n$

$\sum y_n$ serie Riemann $\Rightarrow$

$\Rightarrow\begin{cases}\text{pt. }\beta-1\leq1\Rightarrow\beta\leq2\Rightarrow\sum y_n\text{ diverg}\Rightarrow\sum x_n\text{ diverg.}\\ \text{pt. }\beta-1>1\Rightarrow\beta>2\Rightarrow\sum y_n\text{ converg}\Rightarrow\sum x_n\text{ converg.}\end{cases}$

② $\sum\limits_{n\geq1}\operatorname{tg}^n\left(a+\frac{b}{n^2}\right)$ $\quad 0<a<\frac{\pi}{2}\quad b\in\mathbb{R}$

Crit. Răd $\Rightarrow\lim\limits_{n\to\infty}\sqrt[n]{x_n}=\lim\limits_{n\to\infty}\operatorname{tg}\left(a+\frac{b}{n^2}\right)=\operatorname{tg}a$

I pt. $a\in(0;\frac{\pi}{4})\Rightarrow\operatorname{tg}a\in(0,1)\Rightarrow\lim\limits_{n\to\infty}\sqrt[n]{x_n}<1\Rightarrow\sum x_n$ conv.

II pt. $a\in(\frac{\pi}{4};\frac{\pi}{2})\Rightarrow\operatorname{tg}a\in(1,+\infty)\Rightarrow\lim\limits_{n\to\infty}\sqrt[n]{x_n}>1\Rightarrow\sum x_n$ diverg

III pt. $a=\frac{\pi}{4}\Rightarrow\sum\limits_{n\geq1}\operatorname{tg}^n\left(\frac{\pi}{4}+\frac{b}{n^2}\right)$

$\lim\limits_{n\to\infty}x_n=\lim\limits_{n\to\infty}\operatorname{tg}^n\left(\frac{\pi}{4}+\frac{b}{n^2}\right)=\lim\limits_{n\to\infty}\left(\frac{\operatorname{tg}\frac{\pi}{4}+\operatorname{tg}\frac{b}{n^2}}{1-\operatorname{tg}\frac{\pi}{4}\cdot\operatorname{tg}\frac{b}{n^2}}\right)^n=\lim\limits_{n\to\infty}\left(\frac{1+\operatorname{tg}\frac{b}{n^2}}{1-\operatorname{tg}\frac{b}{n^2}}\right)^n=$

$=\lim\limits_{n\to\infty}\left(\frac{1-\operatorname{tg}\frac{b}{n^2}+2\operatorname{tg}\frac{b}{n^2}}{1-\operatorname{tg}\frac{b}{n^2}}\right)^n=\lim\limits_{n\to\infty}\left(1+\frac{2\operatorname{tg}\frac{b}{n^2}}{1-\operatorname{tg}\frac{b}{n^2}}\right)^n=e^{\lim\limits_{n\to\infty}\frac{2n\operatorname{tg}\frac{b}{n^2}}{1-\operatorname{tg}\frac{b}{n^2}}}$

$=e^{\lim\limits_{n\to\infty}\left(\underset{\downarrow\atop 2}{\frac{2}{1-\operatorname{tg}\frac{b}{n^2}}}\cdot\underset{\downarrow\atop 1}{\frac{\operatorname{tg}\frac{b}{n^2}}{\frac{b}{n^2}}}\cdot\underset{\downarrow\atop 0}{\frac{b}{n}}\right)}=e^{2\cdot1\cdot0}=e^0=1\xRightarrow[\text{nec.}]{\text{Crit.}}\sum\limits_{n\geq1}x_n$ diverg.

③ $\sum\limits_{n\geq1}\int_0^{1/n}\frac{\sin^3x}{1+x^4}dx$

$\sin x\leq x\quad\forall x\in[0;+\infty)\Rightarrow\sin^3x<x^3\Rightarrow\frac{\sin^3x}{1+x^4}\leq\frac{x^3}{1+x^4}\Rightarrow$

$\Rightarrow\int_0^{1/n}\frac{\sin^3x}{1+x^4}dx\leq\int_0^{1/n}\frac{x^3}{1+x^4}dx$

Notăm $x^4=t$ $\quad\int\frac{x^3}{1+x^4}dx=\frac14\int\frac{1}{1+t}dt=\frac14\ln(1+t)+C$

$4x^3dx=dt$

$\Rightarrow\int_0^{1/n}\frac{\sin^3x}{1+x^4}dx\leq\frac14\ln(1+x^4)\Big|_0^{1/n}\Rightarrow\int_0^{1/n}\frac{\sin^3x}{1+x^4}dx\leq\frac14\ln\left(1+\frac{1}{n^4}\right)\leq\frac14\frac{e}{n^4}$

$\ln\left(1+\frac{1}{n^4}\right)^{n^4}\leq e\Rightarrow\ln\left(1+\frac{1}{n^4}\right)\leq\frac{e}{n^4}$

$\frac{e}{4}\sum\frac{1}{n^4}$ converg.

(Serie Riemann cu $\alpha>1$)

$\xRightarrow[\text{I}]{\text{Crit.}}\sum\limits_{n\geq1}\int_0^{1/n}\frac{\sin^3x}{1+x^4}dx$ convergentă

④ $\sum\limits_{n\geq1}\frac{\sqrt{n^2+1}\,(n+1)^\alpha}{\sqrt[3]{n^3+n+1}\,(n+2)}\cdot\ln n$ $\quad\alpha\in\mathbb{R}$ $\quad\frac{\cancel{n}\cdot n^\alpha}{\cancel{n}\cdot n}=\frac{1}{n^{1-\alpha}}$

(prin crit. rap $\to$ lim va fi 1)

fie $y_n=\frac{1}{n^{1-\alpha}}\cdot\ln n$

Crit III $\lim\limits_{n\to\infty}\frac{x_n}{y_n}=\lim\limits_{n\to\infty}\frac{\sqrt{n^2+1}\,(n+1)^\alpha}{\sqrt[3]{n^3+n+1}\,(n+2)}\cdot\cancel{\ln n}\cdot n^{1-\alpha}\cdot\frac{1}{\cancel{\ln n}}=$

$=\lim\limits_{n\to\infty}\frac{\cancel{n^{\alpha+1}}\left(1+\frac1n\right)^\alpha\sqrt{1+\frac{1}{n^2}}}{\cancel{n^2}\left(1+\frac2n\right)\sqrt[3]{1+\frac{1}{n^2}+\frac{1}{n^3}}}\cdot\cancel{n^{1-\alpha}}=\frac{1\cdot1}{1\cdot1}=1\in(0,+\infty)\Rightarrow$

$\Rightarrow\sum\limits_{n\geq1}x_n\sim\sum\limits_{n\geq1}y_n$

Crit. Log $\lim\limits_{n\to\infty}\frac{\ln\frac{1}{y_n}}{\ln n}=\lim\limits_{n\to\infty}\frac{\ln\frac{n^{1-\alpha}}{\ln n}}{\ln n}=\lim\limits_{n\to\infty}\frac{(1-\alpha)\ln n-\ln(\ln n)}{\ln n}$

$=\lim\limits_{n\to\infty}\frac{(1-\alpha)\ln(n)-\ln(\ln n)}{\ln n}=1-\alpha-0=1-\alpha$ $\quad\frac{\ln(\ln n)}{\ln n}\to0$

I Pt $1-\alpha>1\Rightarrow\alpha<0\Rightarrow\sum\limits_{n\geq1}y_n$ conv. $\Rightarrow\sum\limits_{n\geq1}x_n$ conv.

II Pt $1-\alpha<1\Rightarrow\alpha>0\Rightarrow\sum\limits_{n\geq1}y_n$ diverg $\Rightarrow\sum\limits_{n\geq1}x_n$ div.

III Pt $\alpha=0$ atunci $\sum\limits_{n\geq1}y_n=\sum\limits_{n\geq1}\frac{\ln n}{n}$

$\frac{\ln n}{n}>\frac1n$ $\quad\xRightarrow[\text{I}]{\text{Crit.}}\sum\limits_{n\geq1}y_n$ diverg

$\sum\limits_{n\geq1}\frac1n$ diverg. $\quad\Rightarrow\sum\limits_{n\geq1}x_n$ diverg.

⑤ $\sum\limits_{n\geq1}\frac{n^3(-2)^n+5^{2n}\cdot n\sqrt{n}}{n^3\cdot5^{2n+1}}=\sum\limits_{n\geq1}z_n$

$\frac{n^3(-2)^n+5^{2n}n\sqrt n}{n^3\,5^{2n+1}}=\frac{\cancel{n^3}(-2)^n}{\cancel{n^3}\,5^{2n+1}}+\frac{5^{2n}n\sqrt n}{n^3\,5^{2n+1}}=\frac{(-2)^n}{5^{2n+1}}+\frac{1}{5\cdot n^{\frac32}}$

$\sum\limits_{n\geq1}x_n=\sum\limits_{n\geq1}\frac{(-2)^n}{5^{2n+1}}=\sum\limits_{n\geq1}\frac{(-1)^n}{5}\left(\frac{2}{25}\right)^n$ $\quad\xRightarrow[\text{Leibniz}]{\text{Crit.}}\sum\limits_{n\geq1}x_n$ converg.

$\alpha_n=\left(\frac{2}{25}\right)^n$ atunci $\alpha_n\downarrow$

$\alpha_n>0$

$\alpha_n\to0$

$\sum\limits_{n\geq1}y_n=\sum\limits_{n\geq1}\frac{1}{5\cdot n^{\frac32}}\sim\sum\limits_{n\geq1}\frac{1}{n^{\frac32}}$ $\quad\Rightarrow\sum\limits_{n\geq1}y_n$ converg.

$\alpha=\frac32>1\Rightarrow\sum\limits_{n\geq1}\frac{1}{n^{3/2}}$ Serie Riemann converg.

$\sum\limits_{n\geq1}z_n=\sum\limits_{n\geq1}x_n+\sum\limits_{n\geq1}y_n$ $\quad\Rightarrow\sum\limits_{n\geq1}z_n$ converg.

$\sum\limits_{n\geq1}x_n$ şi $\sum\limits_{n\geq1}y_n$ converg.

⑥ $\sum\limits_{n\geq1}\frac{\cos nx}{n^p}$ $\quad x\in\mathbb{R}$ şi $p>0$

$\sum\limits_{n\geq1}\frac{\cos nx}{n^p}=\sum\limits_{n\geq1}\alpha_n$

Fie $x_n=\frac{1}{n^p}$ atunci $\left.\begin{array}{l}x_n\to0\\ x_n\downarrow\\ x_n>0\end{array}\right\}$ (I)

$y_n=\cos(nx)$

$S_n=\sum\limits_{k=1}^n\cos(kx)=\cos x+\cos2x+\dots+\cos(nx)\ \Big|\cdot\sin\frac{x}{2}$

$\sin\frac{x}{2}\cdot S_n=\sum\limits_{k=1}^n\cos kx\cdot\sin\left(\frac{x}{2}\right)=\sum\limits_{k=1}^n\frac12\left[\sin\frac{(2k+1)x}{2}-\sin\frac{(2k-1)x}{2}\right]$

$\sin\frac{x}{2}\cdot S_n=\frac12\left[\sin\left(\cancel{\tfrac32x}\right)-\sin\left(\tfrac12x\right)+\sin\left(\cancel{\tfrac52x}\right)-\sin\left(\cancel{\tfrac32x}\right)+\dots+\sin\frac{(2n+1)x}{2}-\sin\cancel{\frac{(2n-1)x}{2}}\right]$

$\sin\frac{x}{2}\cdot S_n=\frac12\left[\sin\frac{(2n+1)x}{2}-\sin\left(\frac12x\right)\right]$

$\sin\frac{x}{2}\cdot S_n=\frac12\cdot2\sin\left(\frac{nx}{2}\right)\cos\frac{(n+1)x}{2}\Rightarrow S_n=\frac{\sin\left(\frac{nx}{2}\right)\cos\frac{(n+1)x}{2}}{\sin\left(\frac{x}{2}\right)}$

$|S_n|=\left|\frac{\sin\left(\frac{nx}{2}\right)\cos\frac{(n+1)x}{2}}{\sin\left(\frac{x}{2}\right)}\right|\leq\frac{1}{\left|\sin\left(\frac{x}{2}\right)\right|}\Rightarrow S_n$ mărginit (II)

(I);(II) $\xRightarrow[\text{Dirichlet}]{\text{Abel}}\sum\limits_{n\geq1}\alpha_n$ converg.

⑦ $\sum\limits_{n\geq1}n^a\ln\left(1+\frac{1}{n^b}\right)$ $\quad a\in\mathbb{R}$ şi $b>0$

Crit III fie $y_n=\frac{n^a}{n^b}$

$\lim\limits_{n\to\infty}\frac{x_n}{y_n}=\lim\limits_{n\to\infty}\frac{n^a\ln\left(1+\frac{1}{n^b}\right)}{\frac{n^a}{n^b}}=\lim\limits_{n\to\infty}\frac{\ln\left(1+\frac{1}{n^b}\right)}{\frac{1}{n^b}}=$

$=1\in(0,+\infty)\Rightarrow\sum\limits_{n\geq1}x_n\sim\sum\limits_{n\geq1}y_n$

Seria $\sum\limits_{n\geq1}y_n=\sum\limits_{n\geq1}\frac{1}{n^{b-a}}$

pt. $b-a>1\Rightarrow\sum\limits_{n\geq1}y_n$ conv. $\Rightarrow\sum\limits_{n\geq1}x_n$ conv.

pt. $b-a\leq1\Rightarrow\sum\limits_{n\geq1}y_n$ diverg $\Rightarrow\sum\limits_{n\geq1}x_n$ div.

⑧ $\sum\limits_{n\geq1}\frac{\ln(n+1)-\ln n}{\ln^2n}=\sum\limits_{n\geq1}\frac{\ln\frac{n+1}{n}}{\ln^2n}=\sum\limits_{n\geq1}\frac{\ln\left(1+\frac1n\right)}{\ln^2n}=\sum\limits_{n\geq1}x_n$

$\left(1+\frac1n\right)^n<e<\left(1+\frac1n\right)^{n+1}\Rightarrow\ln\left(1+\frac1n\right)^n<1<\ln\left(1+\frac1n\right)^{n+1}$

$\Rightarrow n\ln\left(1+\frac1n\right)<1<(n+1)\ln\left(1+\frac1n\right)\Rightarrow n<\frac{1}{\ln\left(1+\frac1n\right)}<n+1$

$\Rightarrow\frac1n>\ln\left(1+\frac1n\right)>\frac{1}{n+1}\Rightarrow\frac{1}{n\ln^2n}>\frac{\ln\left(1+\frac1n\right)}{\ln^2n}>\frac{1}{(n+1)\ln^2n}$

fie $f:[2,+\infty)\to(0,+\infty)$ $\quad f(x)=\frac{1}{x\ln^2x}$ descresc.

Crit. Int. $\Rightarrow a_n=\int_2^n\frac{1}{x\ln^2x}dx$ $\quad\Rightarrow a_n=\int_{\ln2}^{\ln n}\frac{1}{t^2}dt=\frac{-1}{t}\Big|_{\ln2}^{\ln n}$

Notăm $\ln x=t$ $\quad x=2\Rightarrow t=\ln2$

$\frac1x dx=dt$ $\quad x=n\Rightarrow t=\ln n$

$\Rightarrow a_n=-\frac{1}{\ln n}+\frac{1}{\ln2}$

$a_n=\frac{1}{\ln2}-\frac{1}{\ln n}$

$\lim\limits_{n\to\infty}a_n=\lim\limits_{n\to\infty}\frac{1}{\ln2}-\frac{1}{\ln n}=\frac{1}{\ln2}\Rightarrow a_n$ convergent $\Rightarrow$

$\Rightarrow\sum\limits_{n\geq1}f(n)$ convergent $\quad\Rightarrow\sum\limits_{n\geq1}x_n$ convergentă $\Rightarrow$

$f(n)>x_n$

$\Rightarrow\sum\limits_{n\geq1}\frac{\ln(n+1)-\ln n}{\ln^2n}$ converg.

⑨ $\sum\limits_{n\geq1}(-1)^{n+1}\frac{(n+1)^{n+1}}{n^{n+2}}=\sum\limits_{n\geq1}x_n$

(AC): $\sum\limits_{n\geq1}\left|(-1)^{n+1}\frac{(n+1)^{n+1}}{n^{n+2}}\right|=\sum\limits_{n\geq1}\frac{(n+1)^{n+1}}{n^{n+2}}$

Crit. Raport. $\Rightarrow\lim\limits_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim\limits_{n\to\infty}\frac{(n+2)^{n+2}}{(n+1)^{n+3}}\cdot\frac{n^{n+2}}{(n+1)^{n+1}}=$

$=\lim\limits_{n\to\infty}\frac{\cancel{n^{2n+4}}\left(1+\frac2n\right)^{n+2}}{\cancel{n^{2n+4}}\left(1+\frac1n\right)^{n+3}\cdot\left(1+\frac1n\right)^{n+1}}=\frac{1+0}{(1+0)(1+0)}=1$

fie $y_n=\frac1n$

Crit. III $\Rightarrow\lim\limits_{n\to\infty}\frac{|x_n|}{y_n}=\lim\limits_{n\to\infty}\frac{(n+1)^{n+1}}{n^{n+2}\cdot\frac1n}=\lim\limits_{n\to\infty}\frac{\cancel{n^{n+1}}\left(1+\frac1n\right)^{n+1}}{\cancel{n^{n+1}}}$

$=\lim\limits_{n\to\infty}\left(1+\frac1n\right)^{n+1}=e\in(0,+\infty)\Rightarrow\sum\limits_{n\geq1}|x_n|\sim\sum\limits_{n\geq1}y_n$ $\Rightarrow$

$\sum y_n=\sum\frac1n$ diverg.

$\Rightarrow\sum\limits_{n\geq1}|x_n|$ diverg $\Rightarrow\sum\limits_{n\geq1}x_n$ nu e absolut converg.

(C): fie $\alpha_n=\frac{(n+1)^{n+1}}{n^{n+2}}$ $\quad\alpha_n\geq0$ (I)

$\frac{\alpha_{n+1}}{\alpha_n}=\frac{(n+2)^{n+2}}{(n+1)^{n+3}}\cdot\frac{n^{n+2}}{(n+1)^{n+1}}=\frac{[n(n+2)]^{n+2}}{(n+1)^{2n+4}}=\left(\frac{n^2+2n}{n^2+2n+1}\right)^{n+2}$

$=\left(1-\frac{1}{n^2+2n+1}\right)^{n+2}<1\Rightarrow\alpha_n\downarrow$ (II)

$\lim\limits_{n\to\infty}\alpha_n=\lim\limits_{n\to\infty}\frac{(n+1)^{n+1}}{n^{n+2}}=\lim\limits_{n\to\infty}\frac{\cancel{n^{n+1}}\left(1+\frac1n\right)^{n+1}}{\cancel{n^{n+2}}\,n}=\frac{1^\infty}{\infty}=0$ (III)

(I);(II);(III) $\xRightarrow[\text{Leibn.}]{\text{Crit.}}\sum\limits_{n\geq1}x_n$ converg

⑩ $\sum\limits_{n\geq1}\frac{(-1)^n\,n^\alpha}{\sqrt[3]{n^2+2n+5}-\sqrt[3]{n^2}}$ $\quad\alpha\in\mathbb{R}$

$\sum\limits_{n\geq1}x_n=\sum\limits_{n\geq1}(-1)^n\underbrace{\frac{n^\alpha\left(\sqrt[3]{(n^2+2n+5)^2}+\sqrt[3]{n^4+2n^3+5n^2}+\sqrt[3]{n^4}\right)}{\cancel{n^2}+2n+5-\cancel{n^2}}}_{\alpha_n}$ $\quad\frac{n^\alpha\cdot n\sqrt[3]{n}}{n^1}$

(AC) $\sum\limits_{n\geq1}|x_n|=\sum\limits_{n\geq1}\frac{n^\alpha n\sqrt[3]{n}\left(\sqrt[3]{\left(1+\frac2n+\frac{5}{n^2}\right)^2}+\sqrt[3]{1+\frac2n+\frac{5}{n^2}}+1\right)}{2n+5}$

fie $y_n=\frac{1}{n^{\cancel{1}-\alpha-\cancel{1}-\frac13}}=\frac{1}{n^{-\alpha-\frac13}}=n^{\alpha+\frac13}$

Crit. III $\Rightarrow\lim\limits_{n\to\infty}\frac{|x_n|}{y_n}=\lim\limits_{n\to\infty}\frac{\overset{n}{\cancel{n^{\alpha+1+\frac13}}}\cdot3}{(2n+5)\cancel{n^{\alpha+\frac13}}}=$

$=\frac32\in(0,+\infty)\Rightarrow\sum|x_n|\sim\sum y_n$

$y_n$ serie Riemann $\Rightarrow$

I pt $-\alpha-\frac13>1\Rightarrow\alpha<-\frac43\Rightarrow\sum y_n$ conv. $\Rightarrow\sum|x_n|$ conv.

$\Rightarrow\sum x_n$ absolut conv. $\Rightarrow\sum x_n$ conv.

II pt $-\alpha-\frac13\leq1\Rightarrow\alpha\geq-\frac43\Rightarrow\sum y_n$ div. $\Rightarrow\sum|x_n|$ div.

$\Rightarrow\sum x_n$ nu e AC

(C) $-\alpha-\frac13\leq1\Rightarrow-\alpha\leq\frac43\Rightarrow\alpha\geq-\frac43\Rightarrow\alpha+\frac43\geq0$

III pt. $\alpha+\frac43\in[0;1)\Rightarrow1\geq-\alpha-\frac13>0\Rightarrow\alpha_n\downarrow$

$\lim\limits_{n\to\infty}\alpha_n=0$ $\quad\xRightarrow[\text{Leibniz}]{\text{Crit.}}$

$\alpha_n>0$

$\Rightarrow\sum\limits_{n\geq1}(-1)^n\alpha_n$ convergentă $\Rightarrow\sum\limits_{n\geq1}x_n$ convergentă

IV pt. $\alpha+\frac43\in[1,+\infty)\Rightarrow\nexists\lim\limits_{n\to\infty}x_n\Rightarrow\sum\limits_{n\geq1}x_n$ diverg.

⑪ $\sum\limits_{n\geq1}(-1)^n\cdot\frac{2^n+\operatorname{sh}n}{3^n}$ $\quad\operatorname{sh}n=\frac{e^n-e^{-n}}{2}=\frac{e^n-\frac{1}{e^n}}{2}$

$\operatorname{sh}n=\frac{e^{2n}-1}{2e^n}$

(AC): $\sum\limits_{n\geq1}|x_n|=\sum\limits_{n\geq1}\frac{2^n+\frac{e^{2n}-1}{2e^n}}{3^n}=\sum\limits_{n\geq1}\frac{2^{n+1}e^n+e^{2n}-1}{2\cdot(3e)^n}=$

$=\underset{\downarrow\atop C}{\sum\limits_{n\geq1}\frac{2^n}{3^n}}+\underset{\downarrow\atop C}{\sum\limits_{n\geq1}\frac12\cdot\frac{e^n}{3^n}}-\underset{\downarrow\atop C}{\sum\limits_{n\geq1}\frac12\cdot\frac{1}{(3e)^n}}\Longrightarrow$

(pt. că sunt serii geometrice cu $|q|<1$)

$\Longrightarrow\sum\limits_{n\geq1}x_n$ este absolut converg. $\Rightarrow$

$\Rightarrow\sum\limits_{n\geq1}x_n$ converg.

⑫ $\sum\limits_{n\geq1}(-1)^n\frac{\sqrt{n+1}}{(n^2+n+1)^\alpha}\cdot\frac{1}{\ln^\beta(n+2)}$ ; $\alpha\in\mathbb{R},\ \beta>0$

(AC) $\sum\limits_{n\geq1}|x_n|=\sum\limits_{n\geq1}\frac{\sqrt{n+1}}{(n^2+n+1)^\alpha\ln^\beta(n+2)}$ $\quad\frac{n^{\frac12}}{n^{2\alpha}\cdot\ln^\beta(n+2)}$

Fie $y_n=\frac{1}{n^{2\alpha-1/2}\cdot\ln^\beta(n+2)}$

Crit III $\Rightarrow\lim\limits_{n\to\infty}\frac{|x_n|}{y_n}=\lim\limits_{n\to\infty}\frac{\sqrt{n+1}\cdot n^{2\alpha-1/2}\cancel{\ln^\beta(n+2)}}{(n^2+n+1)^\alpha\cancel{\ln^\beta(n+2)}}=$

$=\lim\limits_{n\to\infty}\frac{\cancel{n^{\frac12}}\cdot\cancel{n^{2\alpha-1/2}}\sqrt{1+\frac1n}}{\cancel{n^{2\alpha}}\left(1+\frac1n+\frac{1}{n^2}\right)^\alpha}=\frac11=1\in(0,+\infty)\Rightarrow$

$\Rightarrow\sum\limits_{n\geq1}|x_n|\sim\sum\limits_{n\geq1}y_n$

Crit. log $\lim\limits_{n\to\infty}\frac{\ln\frac{1}{y_n}}{\ln n}=\lim\limits_{n\to\infty}\frac{\ln\left(n^{2\alpha-1/2}\cdot\ln^\beta(n+2)\right)}{\ln n}=$

$=\lim\limits_{n\to\infty}\frac{\left(2\alpha-\frac12\right)\ln n+\beta\ln(\ln(n+2))}{\ln n}=2\alpha-\frac12$

I Pt. $2\alpha-\frac12>1\Rightarrow\alpha>\frac34\Rightarrow\sum\limits_{n=1}^\infty y_n$ conv. $\Rightarrow\sum\limits_{n=1}^\infty|x_n|$ conv.

$\Rightarrow\sum\limits_{n=1}^\infty x_n$ absol. converg. $\Rightarrow\sum\limits_{n=1}^\infty x_n$ conv.

II Pt. $2\alpha-\frac12<1\Rightarrow\alpha<\frac34\Rightarrow\sum\limits_{n=1}^\infty y_n$ diverg $\Rightarrow\sum\limits_{n=1}^\infty|x_n|$ diverg

$\alpha=\frac34$

$\Rightarrow\sum\limits_{n=1}^\infty|x_n|$ nu e abs. converg. abs. conv

III Pt. $2\alpha-\frac12=1\Rightarrow\alpha=\frac34\Rightarrow\sum\limits_{n\geq1}y_n=\sum\limits_{n\geq1}\frac{1}{n\ln^\beta(n+2)}$

Crit. integ. $f:(0;+\infty)\to[0;+\infty)$ cu $f(x)=\frac{1}{x\ln^\beta(x+2)}$

$\frac{1}{x\ln^\beta(x+2)}<\frac{1}{x\ln^\beta x}\Rightarrow\int_2^n f(x)dx<\int_2^n\frac{1}{x\ln^\beta x}dx$ $\quad f(x)\downarrow$

$\int_2^n\frac{1}{x\ln^\beta x}dx=$

conv? pt. $\boxed{\alpha<\frac34}$

(C): $\sum\limits_{n\geq1}x_n=\sum\limits_{n\geq1}(-1)^n\frac{\sqrt{n+1}}{(n^2+n+1)^\alpha}\cdot\frac{1}{\ln^\beta(n+2)}$

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⑬ $\sum\limits_{n\geq1}\sin\left(\sqrt{n^2+a^2}\cdot\pi\right)=\sum\limits_{n\geq1}x_n$

$x_n=\sin\left(\sqrt{n^2+a^2}\cdot\pi\right)=\sin\left(\sqrt{n^2+a^2}\cdot\pi-n\pi+n\pi\right)$

$=\sin\left(\sqrt{n^2+a^2}\cdot\pi-n\pi\right)\cos n\pi+\sin n\pi\cos\left(\sqrt{n^2+a^2}\,\pi-n\pi\right)$

$=\sin\underset{\frac{ct}{\infty}=0}{\left(\frac{(\cancel{n^2}+a^2)\pi^2-\cancel{n^2\pi^2}}{\sqrt{n^2+a^2}\,\pi+n\pi}\right)}\underset{(-1)^n}{\cos n\pi}+\underset{0}{\sin n\pi}\cos\underset{\frac{ct}{\infty}=0}{\left(\frac{(\cancel{n^2}+a^2)\pi^2-\cancel{n^2\pi^2}}{\sqrt{n^2+a^2}\,\pi+n\pi}\right)}$

$\sum\limits_{n\geq1}x_n=\sum\limits_{n\geq1}(-1)^n\cdot\underbrace{\sin\left(\frac{a^2\pi^2}{\sqrt{n^2+a^2}\cdot\pi+n\pi}\right)}_{\alpha_n}$

$\lim\limits_{n\to\infty}\alpha_n=0$

$\alpha_n\downarrow$ pt. că $\frac{a^2\pi^2}{\sqrt{n^2+a^2}\cdot\pi+n\pi}\downarrow$ $\quad\xRightarrow[\text{Leibniz}]{\text{Crit.}}\sum\limits_{n\geq1}(-1)^n\alpha_n$ converg.

$\alpha_n>0$ $\quad\left(\sum\limits_{n\geq1}x_n\text{ converg.}\right)$

$\sum\left|(-1)^n\alpha_n\right|=\sum\alpha_n\sim\sum\frac{a^2\pi^2}{\sqrt{n^2+a^2}\,\pi+n\pi}\sim\sum\frac1n$ diverg.

(din crit III $\lim\limits_{n\to\infty}\frac{\sin x}{x}=1\in(0;+\infty)$)

$\Rightarrow\sum\limits_{n\geq1}x_n$ nu e absolut convergentă